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Department of Electronics and Communication Engineering

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Course Code: ECE 2104

Course Title: Digital Electronics and Logic Circuits Laboratory

OPEN ENDED LAB

Aim:

To design and implement the solution to the problem statement 1, given as open ended lab.

Problem Statement 1 :

Design a digital system that compares two 4-bit binary numbers and displays the comparison result in a user-friendly way. The system will take two 4-bit inputs (A and B) and determine whether A is greater than, equal to, or less than B. Based on the comparison, the output will be generated accordingly.

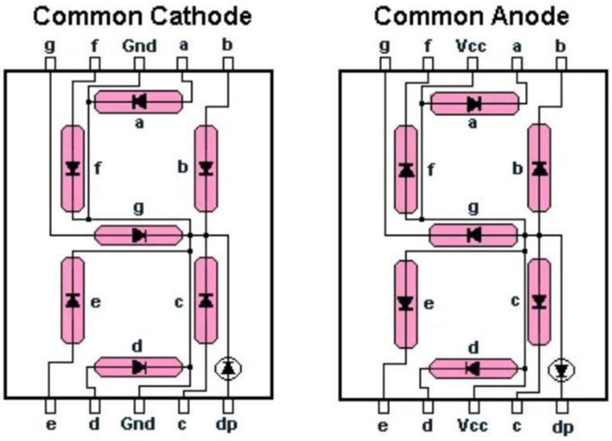
A seven segment display will be used to indicate the result clearly by showing a specific symbol or number representing the condition (A>B, A=B, or A<B). The system should ensure accurate comparison using logic gates and proper encoding to drive the display.

The main challenge is to integrate the 4-bit comparator logic with the display unit so that the output is both correct and easily understandable. This system aims to overcome the limitation of simple LED-based outputs by providing a more informative visual representation of the comparison result.

Main Components:

1 . 7-Segment Display:

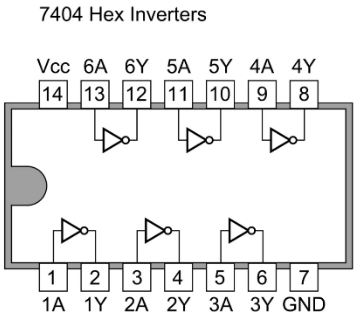
A seven segment display is an electronic device used to show numbers. It has seven LED segments (a–g) that turn on or off to form digits (0–9). It is commonly used in digital systems for simple visual output.
There are two types: common anode and common cathode. It is widely used in clocks, calculators, and digital circuits for displaying results clearly.



SSD Configuration

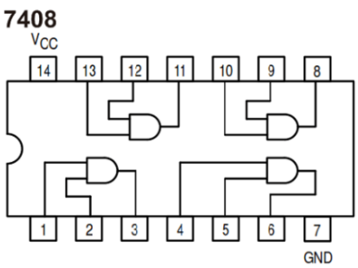
2 . IC 7404 (NOT Gate):

IC 7404 is a digital integrated circuit that contains six independent NOT gates (inverters). Each gate takes one input and produces the opposite (complement) output, meaning it converts logic 1 to 0 and logic 0 to 1. It is commonly used in digital circuits for signal inversion and logic control.



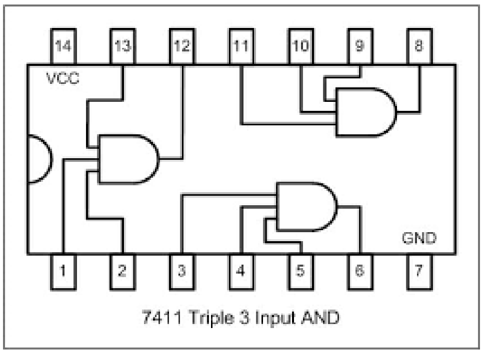
3. IC 7408 (2-input AND):

IC 7408 is a digital integrated circuit that contains four independent 2-input AND gates. Each gate gives a HIGH (1) output only when both inputs are HIGH; otherwise, the output is LOW (0). It is widely used in digital systems for logical multiplication and control operations.



4 . IC 7411 (3-input AND):

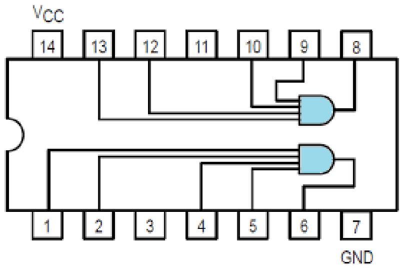
IC 7411 is a digital integrated circuit that contains three independent 3-input AND gates. Each gate produces a HIGH (1) output only when all three inputs are HIGH;



otherwise, the output is LOW (0). It is used in digital circuits where multiple conditions must be satisfied simultaneously.

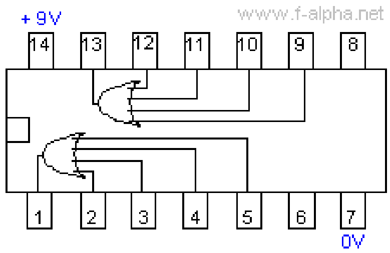
5 . IC 7421 (4-input AND):

IC 7421 is a digital integrated circuit that contains two independent 4-input AND gates. Each gate produces a HIGH (1) output only when all four inputs are HIGH; otherwise, the output is LOW (0). It is used in digital systems where multiple input conditions need to be checked at the same time.



6 . IC 4072 (4-input OR):

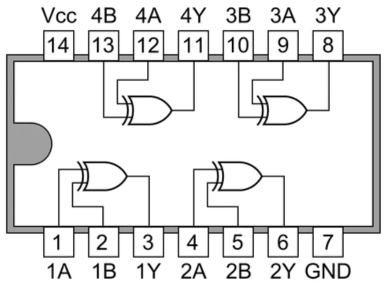
IC 4072 is a CMOS integrated circuit that contains two independent 4-input OR gates. Each gate produces a HIGH (1) output if any one of its inputs is HIGH; the output is LOW (0) only when all inputs are LOW. It is commonly used in digital circuits for combining multiple input signals.



7 . IC 7486 (XOR Gate):

IC 7486 is a digital integrated circuit that contains four independent 2-input XOR (Exclusive-OR) gates. Each gate produces a HIGH (1) output when the two inputs are different, and a LOW (0) output when the inputs are the same. It is widely used in digital circuits for comparison, parity checking, and arithmetic operations.

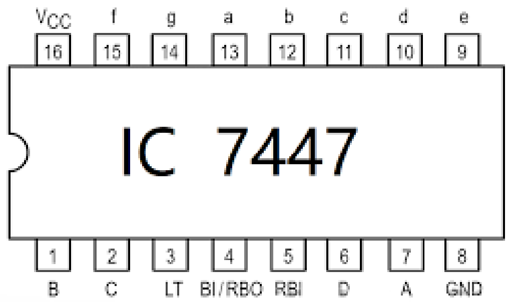
7486 Quad 2-input ExOR Gates



8 . IC 7447 / 4511:

IC 7447 and IC 4511 are decoder/driver ICs used to convert Binary Coded Decimal (BCD) inputs into signals for a seven-segment display. They take a 4-bit BCD input and activate the corresponding segments to display decimal digits (0–9).

IC 7447 is mainly used with common anode displays, while IC 4511 is used with common cathode displays. They are widely used in digital systems for easy numeric display.



Circuit Diagram:

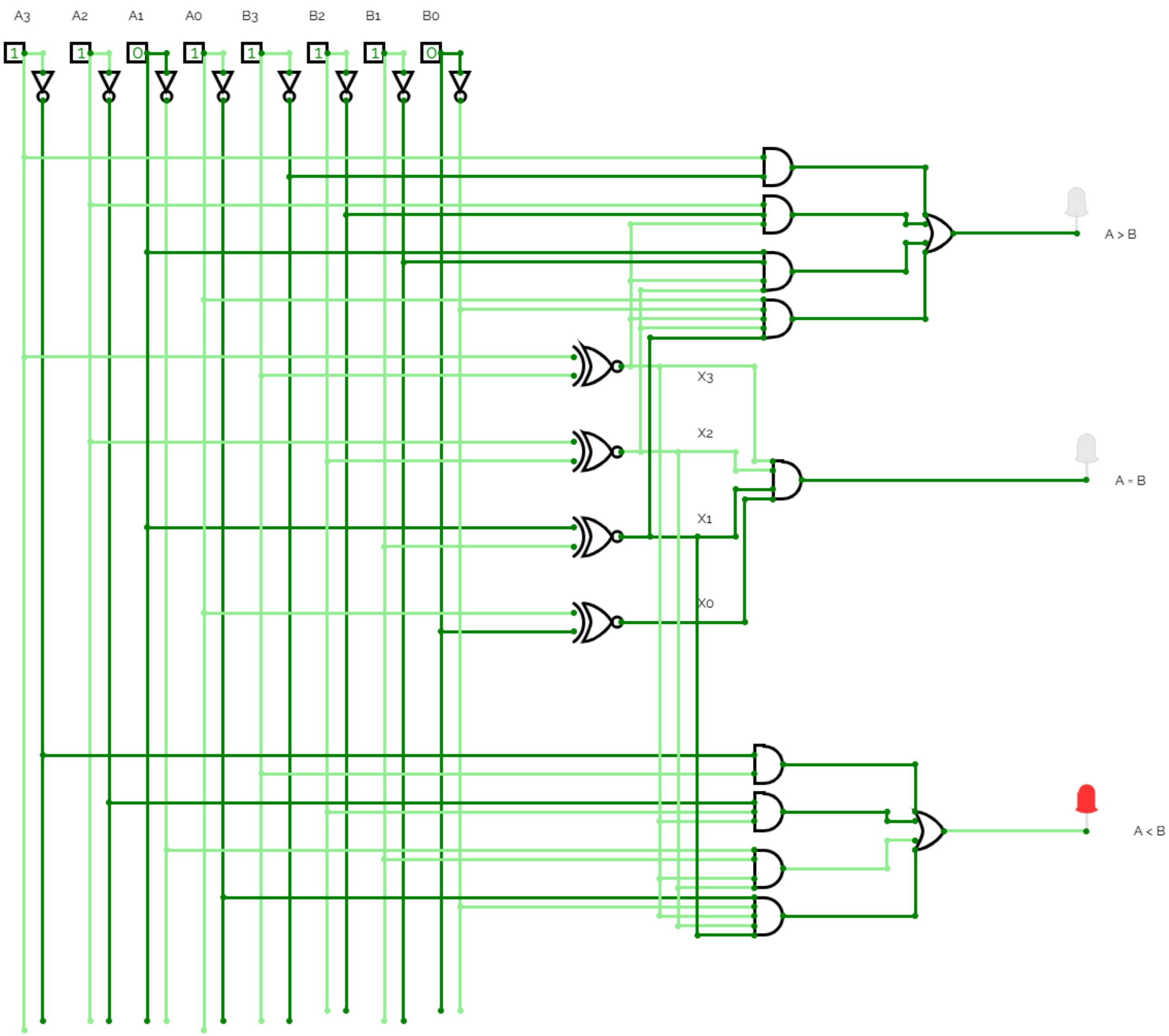


Fig: Circuit Diagram for 4 bit comparator circuit.

Expected outcome :

The proposed project aims to design and implement a functional 4-bit digital comparator circuit using logic gates. The expected outcome is a fully operational system capable of comparing two 4-bit binary inputs (A and B) and accurately determining their relationship.

The circuit will generate three distinct outputs indicating whether A > B, A = B, or A < B. These outputs will be visually represented using LEDs and further displayed through a 7-segment display for clear and user-friendly interpretation. 4

By the completion of the project, the following outcomes are expected:

- Successful construction of the comparator circuit using basic logic gate ICs.
- Accurate comparison of two 4-bit binary numbers in real-time.
- Proper functioning of output indicators (A > B, A = B, A < B).
- Integration of a 7-segment display to present the result in a readable format.
- Improved understanding of digital logic design, gate-level implementation, and practical circuit building.

Overall, the project will demonstrate the practical application of combinational logic design and provide hands-on experience in implementing digital systems using hardware components.

Details of the approach :

The proposed project will be implemented through a structured hardware design methodology using basic digital logic components. The approach focuses on designing a 4-bit comparator at the gate level and verifying its operation through practical implementation.

Initially, the logical relationship between two 4-bit binary numbers (A = A3,A2,A1,A0 and B = B3,B2,B1,B0) will be analyzed. Boolean expressions for the three output conditions (A > B, A = B, and A < B) will be derived using fundamental principles of combinational logic. Equality between corresponding bits will be determined using XOR/XNOR operations, while magnitude comparison will be established using AND, OR, and NOT gate combinations.

In the next step, the derived Boolean expressions will be converted into a complete gate-level circuit diagram. Multi-input logic requirements (such as 3-input, 4-input, and 5-input AND/OR operations) will be implemented using ICs or by combining lower-input gates where necessary.

Following the design phase, the circuit will be physically constructed on a breadboard. Required ICs (such as 7408, 7411, 7421, 7486, 7404, and 4072) will be interconnected using jumper wires. Toggle switches will be used to provide input combinations, and a regulated +5V power supply will be used to operate the circuit.

For output visualization, LEDs will be connected to represent the three comparison conditions. Additionally, a 7-segment display will be integrated using an appropriate decoder/driver IC (such as 7447 or 4511) to present the output in a more user-friendly format.

Finally, the system will be tested with all possible input combinations to verify correctness and reliability. Any discrepancies will be debugged, and necessary adjustments will be made to ensure accurate performance.